

A Unified Green AI Framework for Sustainable and Energy-Efficient Artificial Intelligence Systems

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Abstract—Recent advances in artificial intelligence (AI) have led to significant automation and data-driven decision-making in many areas. But the environmental costs of AI systems in terms of energy consumption, carbon emissions and electronic waste have received far less attention than they deserve. In this study we discuss the environmental implications of contemporary AI and the growing divide between performance-driven development and sustainability-oriented approaches. We propose a unified Green AI framework that integrates energy-efficient model design and optimization techniques such as pruning and quantization. The framework also incorporates transparency mechanisms including carbon reporting and lifecycle assessment to quantify and convey environmental impact across the development process. Our findings suggest that the combination of efficient algorithms and responsible long-term infrastructure can lead to sustainable AI development.

Index Terms—Green AI, sustainable artificial intelligence, energy-efficient models, carbon footprint, lifecycle assessment.

I. INTRODUCTION

Artificial Intelligence (AI) has become one of the most powerful technologies of our time, revolutionizing industries such as healthcare, finance, transportation, education, manufacturing, and scientific research [1], [2]. The fast development of machine learning (ML), deep learning (DL), and large language models (LLMs) has led to significant advances in intelligent automation, predictive analytics, natural language processing, and decision-making [3], [4]. However, as the size and complexity of AI models increase, they also bring serious environmental and sustainability challenges due to their high computational demands, excessive energy consumption, and growing carbon emissions [5], [6].

Studies have shown that large-scale AI systems, especially generative models, require huge computing infrastructure during both training and inference, putting increasing pressure on global energy resources [7], [8]. The environmental footprint

of AI is much more than just computation. Modern AI ecosystems are responsible for carbon emissions throughout the entire lifecycle of computing infrastructure, including hardware manufacturing, cloud operations, model training, deployment, and electronic waste generation [9], [10]. Large models such as GPT-based architectures require huge amounts of electricity, and AI servers and data centers significantly increase energy demand and greenhouse gas emissions [11]. The increasing dependency on dedicated AI hardware also accelerates resource depletion and e-waste generation, making sustainability a truly multi-dimensional challenge [12], [13].

To address these challenges, the concept of Green Artificial Intelligence (Green AI) has emerged as a research direction aimed at developing energy-efficient, carbon-aware, and environmentally responsible AI systems [14]. Green AI aims at minimizing energy consumption and carbon footprint while keeping model performance at an acceptable level [15]. Several sustainability-oriented techniques have been explored in existing works, such as model compression, pruning, quantization, energy-aware scheduling, carbon footprint estimation, and hardware-efficient computing [16], [17]. Monitoring tools such as CarbonTracker and CodeCarbon have been proposed to estimate energy consumption and carbon emissions during training and inference [18].

More recent work has extended Green AI into infrastructure management and intelligent carbon optimization. Multi-agent reinforcement learning frameworks have been proposed for real-time optimization of data center cooling, energy use, and carbon-aware workload scheduling [19]. AI-driven carbon management systems have been introduced for sustainable supply chains and industrial decision-making [20]. Studies have also highlighted the importance of renewable energy integration, AI-driven sustainability innovation, and responsible AI governance for long-term carbon efficiency [21], [22].

Green AI now encompasses cloud optimization, lifecycle carbon assessment, and environmentally responsible deployment methodologies [23].

Despite this progress, existing Green AI research remains fragmented. Most current studies address isolated aspects of sustainability such as energy monitoring, carbon estimation, hardware optimization, or inference efficiency without offering a unified sustainability-oriented architecture [24]. Some approaches only measure emissions during training, while others focus exclusively on inference or hardware-level efficiency [25]. Many existing frameworks also fail to consider the full lifecycle impact of AI systems, including hardware use, deployment conditions, infrastructure sustainability, and end-of-life consequences [26]. The absence of integrated monitoring, optimization, evaluation, and reporting mechanisms makes it difficult to establish transparent and standardized sustainability management [27].

Inspired by these challenges, this paper proposes a *Unified Green AI Framework for Sustainable and Energy-Efficient Artificial Intelligence Systems*. The proposed framework, in opposition to previous frameworks that target individual sustainable components, brings energy and carbon tracking, lifecycle-sustainable assessment, optimization, dual-criteria evaluation, and iterative improvement into a single system. The proposed framework is a feedback-driven closed-loop system that dynamically achieves trade-off between the performance of the AI models and sustainability requirements. The primary contributions of this paper are as follows:

- **Unified Green AI Framework:** Single architecture that combines energy optimization, carbon monitoring, lifecycle sustainability evaluation, dual-criteria evaluation, and iterative improvement.
- **Lifecycle-Aware Sustainability:** The assessment considers the training, inference, hardware utilization, and deployment phases of the AI system.
- **Energy and Carbon Monitoring and Optimization:** The mechanism for continuous tracking of energy and carbon footprints with the simultaneous increase in computational efficiency via pruning and quantization.
- **Dual-Criteria Evaluation:** AI model performance and sustainability are evaluated simultaneously to avoid trade-offs.
- **Iterative Adaptive Optimization:** A feedback-driven iterative adaptive optimization process that ensures sustainability in real-world deployments of AI.

In brief, the current study is an attempt to narrow the divide between the performance and sustainability of artificial intelligence research and development. This study also lays the foundation for future research in developing sustainable and scalable AI solutions.

II. RELATED WORK

A. Literature Review

Bolón-Canedo et al. [18] studied how sustainable AI practices can reduce the environmental footprint of AI systems,

differentiating between Green-in AI and Green-by AI. The paper provided a good conceptual overview but was mainly review-based and did not provide a practical unified implementation framework.

You et al. [6] studied the environmental impacts of large-scale AI server infrastructure and concluded that the growth of AI servers is directly proportional to electricity usage, water consumption, cooling demand, and carbon emissions. However, this study was limited to the United States and did not address optimization at the model level.

Ligozat et al. [9] investigated the hidden environmental costs of AI systems through Life Cycle Assessment (LCA). Their findings showed that AI sustainability encompasses not only electricity use but also data collection, storage, processing, hardware manufacturing, and disposal. Despite the breadth of this analysis, this study did not connect its lifecycle findings to concrete optimization techniques.

De Vries [15] analyzed the growing energy footprint of AI inference using industry data and demonstrated that the inference phase can be responsible for a greater part of an AI system's overall energy footprint than training. However, the study was descriptive and did not provide any suggestion for practical optimization or lifecycle solutions.

Anthony et al. [20] developed CarbonTracker, a carbon footprint estimation and prediction framework for deep learning model training. The above work increased access to carbon accounting but was only limited to training and did not account for inference emissions and end-of-life of hardware.

Mukta et al. [21] developed a comprehensive Green AI framework that included pruning, quantization, efficient algorithms, TPUs and FPGAs usage, and approaches to building sustainable data centers. The above research was highly correlated with the aims of the current research but did not present a concrete implementation pathway or a life-cycle approach.

Luccioni et al. [24] calculated the carbon footprint during the entire life-cycle of the BLOOM language model, which included training, inference, and hardware production. The above study revealed the invisible cost of developing language models. However, this particular research was related only to one language model and could not be generalized.

All the above research proves that Green AI is becoming an increasingly popular and important topic of research. However, at the moment, there is still no scalable framework that combines all the aspects of energy efficiency, carbon accounting, life-cycle analysis, and sustainable infrastructure development within one framework.

B. Gap Analysis

Table I highlights the key deficiencies of the current research efforts in the area of Green AI in different aspects of research. Every effort is certainly valuable in its own way; however, they all share one characteristic – they always look at sustainability through a narrow scope without trying to understand it from the holistic perspective. The majority of them only consider one aspect like, for example, the training phase, the use of the server infrastructure, or efficient inference without working

towards building a more holistic and scalable solution. Many efforts have yet to be fully implemented or even tested which makes the deployment of AI in a sustainable way rather challenging. There is no existing framework that considers all four aspects – optimization, transparency, hardware effect, and carbon-aware evaluation – in one solution.

III. PROPOSED WORK

A. Overview

This part of the essay discusses the design and architecture of the proposed green AI framework. Most researches in literature have focused on one of the two factors of sustainable AI: some models have taken care of carbon emission while others of energy consumption during inference. On the other hand, the proposed framework considers both these factors in one single framework which is built on the idea of environmental consciousness. The main focus of this framework is to strike a good balance between performance of AI models and their sustainability environmentally. There are three important environmental challenges that the current generation AI models face: the first one is the demand of large amounts of energy in both training and inference. The second challenge is the emission of carbon dioxide from high-performance computing systems. And the last one is the generation of electronic waste when manufacturing and disposing of AI equipment. Our framework considers environment in its design phase and not in deployment phase.

B. Goals and Scope of Design

There are four key goals of our proposed framework:

- **Unified Integration:** The framework was built with awareness of the entire lifecycle of the AI including data collection, training, deployment, hardware use and hardware disposal and hence incorporates sustainability into all stages.
- **Lifecycle Awareness:** The proposed framework clearly measures and reports on its environmental impacts using standardized techniques so as to allow comparisons between different AI systems and hold the developer accountable for the environmental cost of developing the AI.
- **Transparency and Accountability:** The proposed framework clearly measures and reports on its environmental impacts using standardized techniques so as to allow comparisons between different AI systems and hold the developer accountable for the environmental cost of developing the AI.
- **Practical Deployability:** The framework is deployable in the real world which has uncertain and scarce resources as opposed to the academic research environment with controlled resources.

The framework captures both training and inference stages of the AI lifecycle. This is important since earlier approaches such as CarbonTracker [20] considered emissions during the training stage while de Vries [15] only analyzed the inference energy footprint but did not provide solutions.

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C. Architectural Concept

Overview of the suggested framework high-level architecture comprises three interrelated levels operating in a cycle of continuous feedback-based approach rather than sequentially. The first level is the **Data and Monitoring Level** which performs the task of collecting and analyzing environmental and operational data including energy consumption, hardware activity and lifecycle data. The continuous monitoring of system operation ensures the continuous access to sustainability-related data at all stages. The second level is the **Optimization and Decision Level**, which is focused on environmental efficiency enhancement of AI models through reducing their computational complexity and memory footprint with the help of high-level optimizations. Finally, the **Evaluation and Reporting Level** performs evaluation of both performance accuracy and environmental impact simultaneously and creates structured sustainability reports according to standardized assessment principles for improving environmental accountability [9], [20].

Suggested three-level architecture of the Unified Green AI framework overview is shown in Fig. 4.

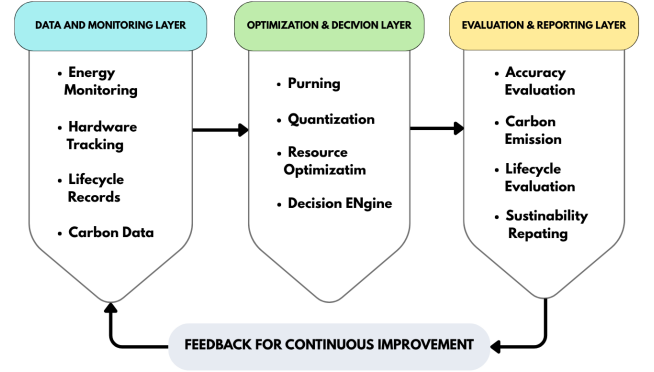


Fig. 1. Three-layer architecture of the Unified Green AI framework that was proposed.

IV. METHODOLOGY

This section describes the detailed implementation of the proposed Green AI framework. The framework is a systematic, closed-loop process of continuously monitoring, evaluating, and improving AI models. The process includes several basic steps: data collection and preparation, building a baseline model, optimization, monitoring, prediction, and iterative refinement.

TABLE I
COMPARISON OF EXISTING GREEN AI STUDIES

Year	Paper	Methodology	Dataset / Case	Limitations (Gap)
2024	A Review of Green AI: Towards a More Sustainable Future [18]	Reviewed Green-in AI and Green-by AI approaches, energy-efficient ML models, and carbon-aware computing.	Sustainable AI systems, ML energy usage, LLM sustainability	Mostly conceptual; no practical unified implementation framework.
2025	Environmental Impact and Net-Zero Pathways for AI Servers in the USA [6]	Analyzed AI server energy, water, and carbon footprints using projection models.	AI server infrastructure across the USA	Focused on server infrastructure only; lacked model-level optimization.
2022	Unraveling the Hidden Environmental Impacts of AI Solutions: LCA of AI [9]	Applied Life Cycle Assessment (LCA) to AI services including hardware, storage, and processing.	AI lifecycle assessment	Did not integrate practical optimization methods such as pruning or quantization.
2023	The Growing Energy Footprint of Artificial Intelligence [15]	Analyzed energy demands of AI inference using industry data and server estimates.	ChatGPT/GPT-based AI inference workloads	Focused on inference energy footprint; did not propose optimization or lifecycle solutions.
2020	CarbonTracker: Tracking and Predicting the Carbon Footprint of Training DL Models [20]	Developed CarbonTracker for monitoring carbon emissions during DL model training.	CNN models, medical image datasets, GPU training	Focused only on training emissions; inference and hardware disposal not considered.
2022	Estimating the Carbon Footprint of BLOOM, a 176B Parameter Language Model [24]	Estimated lifecycle carbon emissions of BLOOM LLM including training, inference, and hardware manufacturing.	BLOOM 176B language model, Hugging Face infrastructure	Focused on a single model; limited generalizability to other architectures and deployment contexts.

A. Data Preparation and Framework Construction

1) *The Framework Pipeline*: The proposed framework is an iterative framework which constantly aims at improving the AI algorithm based on performance and sustainability parameters. Data is collected and processed to train an initial model and then complexity of the algorithm is minimized while computational efficiency is enhanced through the use of optimization algorithms. The optimized model is evaluated based on the mentioned performance and environmental criteria. If the results are not up to the standard, then the model is further trained with new parameters. This cycle repeats until the optimal accuracy and efficiency is achieved.

2) *Data Collection and Preprocessing*: In order to create a representative dataset, various facets of environmental and computational impacts of AI are considered in the form of energy consumption, carbon emissions, hardware usage, lifecycle factors such as e-waste. In order to ensure the precision of data, open-source data repositories and academic literature are used as data sources. Data preprocessing is done prior to the model training stage to improve its quality and computational efficiency.

3) *Developing the Baseline Model*: The artificial neural network (ANN) is employed for developing a baseline model in order to establish initial measures of the performance of the model. No optimization is done while training the model, and the model includes performance measures such as accuracy, loss, and energy consumption.

B. Optimization and Monitoring

1) *Energy and Carbon Tracking*: Energy consumption and carbon footprint are measured in real-time both during the training and inference processes. This is achieved by means of specific tools that measure resource consumption and estimate

their impact on the environment. Thanks to this approach, the framework provides comprehensive sustainability assessment through the whole process.

2) *Model Optimization Techniques*: Model Optimization Methods: The two major methods of optimization pursued are focused on increasing computational efficiency and reducing environmental impact.

a) *Pruning*: In pruning, unnecessary connections are deleted from the neural network. Reducing the number of redundant parameters helps increase computational efficiency and save energy when training and performing inference.

b) *Quantization*: Quantization refers to the procedure of lowering the numerical precision of parameters in a model. In this case, floating point representations are transformed into lower bit values. The consequence is lower memory consumption and higher computational efficiency. After quantization, the model is thoroughly evaluated for any loss of accuracy.

3) *Prediction and Deployment*: After optimization, the algorithm is applied for prediction and testing its use in real life situations. At this point, operational metrics such as latency, power consumption, and efficiency of predictions are calculated. This helps us determine the efficiency and effectiveness of our model in real-life situations, even if the conditions are uncontrolled.

C. Evaluation and Final Output

1) *Evaluation of Performance and Sustainability*: The model evaluation scheme involves a two-criteria scheme for evaluation in terms of performance and sustainability simultaneously. Metrics like accuracy and loss evaluate the model performance whereas sustainability is estimated by evaluating the model in terms of energy consumption, carbon emissions, and resource utilization. Evaluation from two perspectives

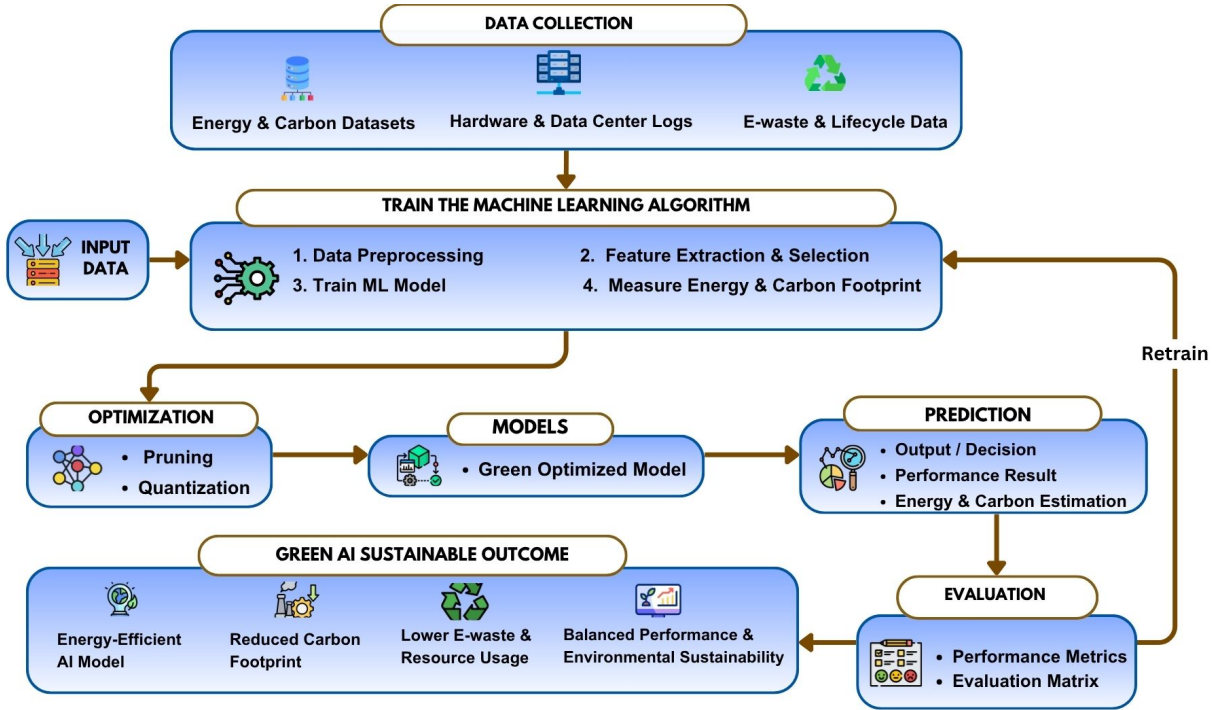


Fig. 2. Proposed Green AI framework workflow.

at the same time helps the framework overcome the usual problem of conflicting goals optimization.

2) *Loopback*: If the evaluation does not provide a desirable result, the framework starts the loopback procedure. The parameters of the model are optimized and then the model is trained again. The procedure repeats itself until the tradeoff between sustainability and performance reaches an acceptable level.

3) *Final Green AI Output*: The result of the framework execution is a Green AI model which offers a tradeoff between small environmental footprint and high performance. The model consumes less energy and emits less carbon dioxide and performs computations more efficiently while still offering good enough accuracy. The full flowchart of the proposed solution is presented in Fig. 2.

V. WORKFLOW EXPLANATION

The workflow of the proposed Green AI model is depicted in Fig. 2. The workflow is categorized into three sub-sections: (A) Model design data acquisition, (B) Model optimization monitoring, and (C) Model evaluation output, creating a closed-loop feedback workflow. Firstly, the data is collected from multiple sources. These sources include energy consumption data sets, hardware data, and lifecycle data such as e-waste data. The above-mentioned data is preprocessed and trained using an Artificial Neural Network (ANN). This ANN acts as a baseline ANN that records performance parameters including accuracy, loss, and energy consumption. Energy consumption and carbon emissions are recorded in real-time

during the training process. After training, the model is further optimized using the process of pruning and quantization. This combination of processes leads to the creation of a more compact and efficient ANN without much sacrifice in accuracy. This optimized ANN is put into use in inference mode, and performance parameters are recorded. The model is then tested using two criteria, which include performance metrics together with sustainability metrics, such as energy consumption, carbon emissions, and resource consumption. In case the outcomes of the tests do not reach the desired level, then a feedback loop will be started to make changes in the parameters of the model. This process is repeated until an adequate balance is reached in terms of both performance and environmental sustainability. Finally, a green AI model will be attained, which has energy efficiency, lower energy consumption, lower carbon emissions, and efficient computing capacity.

VI. EXPERIMENTAL IMPLEMENTATION

A. Experimental Setup

The Green AI framework has been coded using Python in Google Colab platform. The experiments have been done using cloud-based computational resources featuring Intel Xeon CPU and 12 GB of RAM, and NVIDIA Tesla T4 GPU support. The Green AI framework has been validated using publicly available datasets from the Our World in Data (OWID) dataset repository and the UCI Machine Learning Repository. The datasets have been pre-processed through data cleaning, normalization and feature selection. The cleaned

TABLE II
EXPERIMENTAL ENVIRONMENT

Component	Specification
Programming Language	Python 3.x
Development Environment	Google Colab
CPU	Intel Xeon
RAM	12 GB
GPU	NVIDIA Tesla T4
Carbon Monitoring Tool	CodeCarbon
Optimization Techniques	Pruning, Quantization
Dataset Sources	OWID CO ₂ Dataset, UCI Energy Efficiency Dataset

data have been divided into training and test sets in order to conduct model development and evaluation. An Artificial Neural Network (ANN) has been considered as the baseline model to compare with optimization results. The experimental setup and framework implementation of the proposed Green AI approach are shown in Table II.

TABLE III
COMPARISON OF ENERGY EFFICIENCY AND CARBON PREDICTION MODELS

Metric	Energy Model (Best)	Carbon Model (Best)
R^2 Score	0.9323 (Pruned/Quantized)	0.8430 (Pruned)
RMSE	2.4378 (Pruned)	2.5703 (Pruned)
MAE	1.6795 (Quantized)	1.0938 (Baseline)
CO ₂ (kg)	6.58E-7 (Quantized)	6.60E-7 (Pruned)
Latency (ms)	0.95 (Quantized)	2.90 (Pruned)
Sparsity (%)	27.59	28.06

B. Monitoring and Optimization Tools

CodeCarbon has been introduced within the framework to track the effect on the environment during the use of AI models. The system continuously tracked the utilization of computational resources and calculated the associated carbon footprint of the process of training and inference of the model. It allowed for the transparent assessment of indicators related to sustainability during the experiment. Two algorithms for optimization were introduced to increase computational efficiency. Pruning was utilized to exclude unnecessary parameters of neural networks, thereby reducing the complexity and computational resources needed.

C. Dual-Criteria Evaluation Framework

The developed approach has been assessed by using a two-criteria approach, which involved evaluating model performance and sustainability aspects at once. Predictive performance has been estimated by R^2 Score and Root Mean Square Error (RMSE). Sustainability has been evaluated in terms of carbon emissions, inference latency and sparsity of the model. There have been used three types of model in the experiments: ANN model, pruned and quantized models. The metrics gathered have been used to compare the efficiency of optimization approaches in terms of minimizing

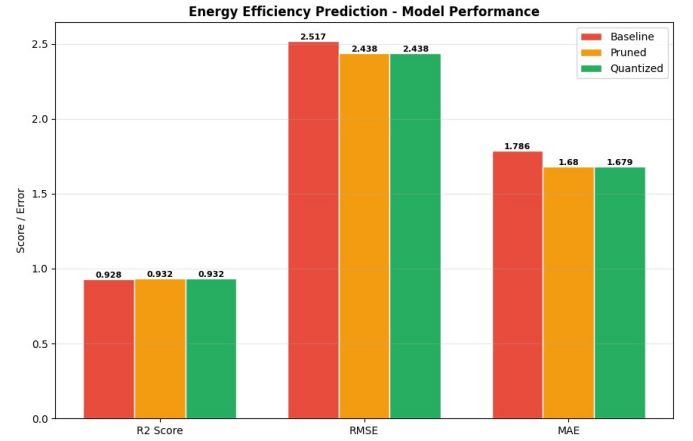


Fig. 3. Comparing performance of baseline, pruned, and quantized models on energy efficiency prediction by using R^2 , RMSE, and MAE.

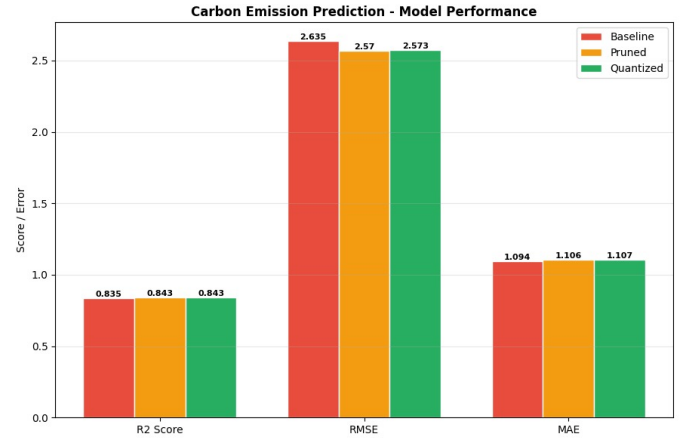


Fig. 4. Comparing performance of baseline, pruned, and quantized models on carbon emission prediction by using R^2 , RMSE, and MAE.

environmental footprint and sustaining predictive performance. Thus, the presented evaluation framework provides a balanced assessment of both the performance and sustainability goals in order to develop sustainable AI systems. In order to evaluate the efficiency of the proposed approach, the best results in terms of performance and sustainability achieved from Energy Efficiency and Carbon Prediction models are compared in Table III.

VII. CONCLUSION

In this work, we propose an integrated Green AI framework in order to address the impact of the environmental footprint of AI technology. Energy measurement, carbon emissions tracking, lifecycle assessment, model optimization, and double-criteria evaluation have been brought together under one framework, which is not only realistic but also practical to implement. The use of strategies such as pruning and quantization in an iterative process with feedback minimizes computational cost and energy use without causing major sacrifices in efficiency. This paper proposes a new Green AI

framework that aims to fill a gap in Green AI research, which to date has mainly concentrated on individual sustainability elements. Through consideration of all training and inference phases along with the use of hardware and lifecycle evaluation, this framework takes the holistic approach towards sustainable development of AI technology that no earlier research has achieved. For future work, there will be efforts made to scale up the framework to larger and different datasets and models and to improve the lifecycle assessment modules. We also plan to explore integration with renewable energy and sustainability report generation on cloud platforms.

REFERENCES

- [1] S. Khan, N. S. Naz, T. Mazhar, M. U. Tariq, T. Shahzad, S. Guizani, and H. Hamam, "Green AI techniques for reducing energy consumption in AI systems," *Neurocomputing*, vol. 616, p. 128671, 2025, doi: 10.1016/j.neucom.2024.128671.
- [2] A. Mina, "Green intelligence: the AI content of green technologies," *Eurasian Business Review*, vol. 15, no. 3, pp. 803–840, Sep. 2025, doi: 10.1007/s40821-024-00288-1.
- [3] X. Wang, "How does artificial intelligence impact green development? Evidence from China," *Sustainability*, vol. 16, no. 3, p. 1260, Feb. 2024, doi: 10.3390/su16031260.
- [4] S. Wen, "Deep learning and artificial intelligence in sustainability: A review of SDGs, renewable energy, and environmental health," *Sustainability*, vol. 15, no. 18, p. 13493, Sep. 2023, doi: 10.3390/su151813493.
- [5] M. Z. Ahmad, J. Rana, M. F. Islam, M. N. I. Jihad, M. Rashed, and M. K. Uddin, "Innovation pathways to carbon efficiency: Disentangling the effects of AI, R&D, and clean energy blessings on U.S. environmental sustainability," *Business Strategy and the Environment*, pp. 1–19, 2025, doi: 10.1002/bse.70748.
- [6] F. You et al., "Environmental impact and net-zero pathways for sustainable artificial intelligence servers in the USA," *Nature Sustainability*, vol. 8, no. 12, pp. 1541–1553, Dec. 2025, doi: 10.1038/s41893-025-01681-y.
- [7] S. Wen, "Deep learning and artificial intelligence in sustainability: A review of SDGs, renewable energy, and environmental health," *Sustainability*, vol. 15, no. 18, p. 13493, Sep. 2023, doi: 10.3390/su151813493.
- [8] Z. Fan, "Sustainable innovation management: Balancing economic growth and environmental responsibility," *Sustainability*, vol. 17, no. 10, p. 4362, 2025, doi: 10.3390/su17104362.
- [9] A.-L. Ligozat, J. Lefèvre, A. Bugeau, and J. Lottier, "Unraveling the hidden environmental impacts of AI solutions for environment life cycle assessment of AI solutions," *Sustainability*, vol. 14, no. 9, p. 5172, Apr. 2022, doi: 10.3390/su14095172.
- [10] R. Li, "Ecological footprints, carbon emissions, and energy transitions: The impact of artificial intelligence (AI)," *Humanities and Social Sciences Communications*, vol. 11, no. 1, p. 1043, Aug. 2024, doi: 10.1057/s41599-024-03520-5.
- [11] B. Bergougui, "Can artificial intelligence technologies advance environmental sustainability? The role of institutional adaptability and skill-biased technological transformation," *Sustainable Development*, vol. 34, no. S2, pp. 222–244, 2026, doi: 10.1002/sd.70296.
- [12] S. Pimenow, O. Pimenowa, and P. Prus, "Challenges of artificial intelligence development in the context of energy consumption and impact on climate change," *Energies*, vol. 17, no. 23, p. 5965, Nov. 2024, doi: 10.3390/en17235965.
- [13] T. Wang, "The impact of China's artificial intelligence development on urban energy efficiency," *Scientific Reports*, vol. 15, no. 1, p. 24129, Jul. 2025, doi: 10.1038/s41598-025-09319-x.
- [14] C. Clemm and K. Wimalawarne, "Towards Green AI: Current status and future research," in *Proc. IEEE Int. Conf.*, 2024, doi: 10.1109/10549890.
- [15] A. de Vries, "The growing energy footprint of artificial intelligence," *Joule*, vol. 7, no. 10, pp. 2191–2194, Oct. 2023, doi: 10.1016/j.joule.2023.09.004.
- [16] J. Huang and S. Gopal, "Artificial intelligence and climate change: Ethical issues," *Journal of Information, Communication and Ethics in Society*, vol. 21, no. 1, pp. 1–16, 2023, doi: 10.1108/JICES-06-2022-0063.
- [17] V. Bolón-Canedo, B. Remeseiro, and A. Alonso-Betanzos, "A review of green artificial intelligence: Towards a more sustainable future," *Neurocomputing*, vol. 599, 2024, doi: 10.1016/j.neucom.2024.128371.
- [18] V. Bolón-Canedo, B. Remeseiro, and A. Alonso-Betanzos, "A review of green artificial intelligence: Towards a more sustainable future," *Neurocomputing*, vol. 599, 2024, doi: 10.1016/j.neucom.2024.128371.
- [19] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, "Green AI," *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, Dec. 2020, doi: 10.1145/3381831.
- [20] L. F. W. Anthony, B. Kanding, and R. Selvan, "CarbonTracker: Tracking and predicting the carbon footprint of training deep learning models," arXiv:2007.03051, Jul. 2020. [Online]. Available: <https://arxiv.org/abs/2007.03051>
- [21] M. S. H. Mukta et al., "Towards sustainable AI: A comprehensive framework for Green AI," *Discover Sustainability*, vol. 5, no. 1, p. 408, Dec. 2024, doi: 10.1007/s43621-024-00641-4.
- [22] I. Schneider, "An introduction to life-cycle emissions of artificial intelligence hardware," *IEEE Intelligent Systems*, vol. 40, no. 5, 2025, doi: 10.1109/MI.2025.11097303.
- [23] T. Tunzina, "GreenAI: A comparative analysis of environmental efficiency in LLM-generated code," *IEEE Transactions*, 2025, doi: 10.1109/11366639.
- [24] A. S. Luccioni, S. Viguier, and A.-L. Ligozat, "Estimating the carbon footprint of BLOOM, a 176B parameter language model," arXiv:2211.02001, Nov. 2022. [Online]. Available: <https://arxiv.org/abs/2211.02001>
- [25] M. A. Uddin et al., "A review of the recent development, challenges, and opportunities of electronic waste (e-waste)," *International Journal of Environmental Science and Technology*, vol. 20, no. 4, pp. 4513–4520, Apr. 2023, doi: 10.1007/s13762-022-04274-w.
- [26] C. Clemm and K. Wimalawarne, "Towards Green AI: Current status and future research," in *Proc. IEEE Int. Conf.*, 2024, doi: 10.1109/10618912.
- [27] A. R. Babu et al., "Carbon footprint reduction for sustainable data centers in real-time," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, pp. 22322–22330, Mar. 2024, doi: 10.1609/aaai.v38i20.30238.